

Kingdom of Saudi Arabia
Ministry of Education
Qassim University
College of Computer
Information Technology Dept.



المملكة العربية السعودية وزارة
التعليم
جامعة القصيم كلية
الحاسب
قسم تقنية المعلومات

IoT-Enabled Solar-Powered Smart Wheelchair

Students:

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|-----------------------------|-----------|
| 1. Mansour Mutlaq Alharbi | 411108002 |
| 2. Meshari Abdullah Alsaegh | 382123968 |
| 3. Saud Faisal Alfadda | 382128151 |

Supervisor:

Dr. Salim El Khediri

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Literature Review :

Solar Powered Wheelchair with Voice Control and Gesture Controller

There has been a growing interest in integrating intelligent technologies into assistive mobility with the aim of facilitating independence in physically disabled persons. The following is the thematic synthesis of existing work and the gaps that the project addresses:

1. Assistive Mobility and Control Systems

The importance of multimodal control in making systems more accessible has received great emphasis in past research. For example, a solar-powered wheelchair controlled by gestures and voice was proposed by Haque et al. (2020) in order to reduce physical effort. Similarly, Madheswari et al. (2017) presented voice control navigation using low-cost hardware, but IoT was not implemented in that paper. The mentioned works show major principles of user-centered interfaces, but the smooth integration of different control modes-manual switch, gesture, voice-is still a lacuna.

This review of gesture recognition technologies by Chen 2018 further validated the reliability of motion sensors, such as GY-61, for human-machine interaction. Although there were also challenges associated with resistance to environmental noise, this agrees with the use of gesture control in the current project but has brought to light the need for robust testing in diverse conditions.

2. Sustainable Energy Solutions

Solar energy has become one of the major alternatives for feeding assistive devices, especially in areas with poor access to electricity. Vijayakumar et al. 2021 presented a solar-powered wheelchair with smart features, outlining the reduced cost of operation when compared with conventional systems. The work, however, established inefficiency in the batteries at low sunlight conditions, a limitation improved in the current proposal through upgrading to lithium-ion batteries.

3. IoT and Health Monitoring

IoT-based health monitoring is key to any form of real-time medical intervention. Baker et al. (2020) discussed IoT in healthcare, especially using ThingSpeak for data analysis. The system here tries to extend this work by integrating heart rate and temperature sensors to allow remote monitoring by caregivers and extends previous works that were centered around mobility aspects only.

4. Technical Implementation and Open-Source Frameworks

Practical guides, such as Monk's 2016 work, *Programming Arduino*, provide the basic knowledge for prototyping motion control systems. Other examples of the validity of modular design come from existing open-source projects which implement, for example, voice-controlled wheelchairs with Arduino, but most projects simply do not explain how they can be scaled, as will be explained in this paper with open-source code and circuit diagrams.

5. Gaps in Existing Research

Limitations to Safety: Many existing systems, including Haque et al. (2020), do not have obstacle avoidance sensors to prevent collisions.

Battery Dependency: Lead-acid batteries (Madheswari et al., 2017) are less efficient than modern alternatives.

User Adaptation: Limited field testing with severely disabled users (Chen, 2018) raises questions about practicality.

6. Contributions of the Current Work

This project addresses these gaps by:

Integrating infrared sensors for obstacle detection.

Proposing lithium-ion batteries for extended lifespan.

Emphasize field testing with a wide range of users.

Pros:

Super Flexible Controls: Melds voice, gesture, and manual switches to enable users to take advantage of which works best.

Solar Power = Eco-Friendly: Saves cash on electricity while being perfect for sunny areas without power outlets.

Health Monitoring for Safety: Allow doctors to verify your heart rate and temperature in comfort.

Low-Cost to Construct: Made of affordable parts like Arduino and able to be integrated into old wheelchairs, rather than buying new ones.

Mobile Application to Control: Ability to drive around using a smartphone application—absolutely futuristic!

Cons:

Shitty Bluetooth Range: Works up to only 10 meters; walls ruin the signal.

Not Good to Go in Bad Weather: Solar panels do not charge well on overcast or rainy days.

No Protection against Collisions: There are no sensors that would stop you from bumping into the wall or into furniture.

Gestures are confusing: it may be difficult for the older users to learn hand motions.

Battery Life Problems: It requires frequent charging when Bluetooth or the app is used too much.

Student's Notes :

Add Safety Sensors: Infrared or ultrasonic sensors that would avoid crashing by the

Upgrade the Battery: Lithium-ion batteries last longer compared to the old lead-acid ones.

Usability Test: Allow disabled users to try if it's practically workable.

Code Sharing: Arduino sketches can be uploaded online for others to help improve the design.

Alternative Approach: In case the design will not work using Bluetooth, then it shall have manual controls working perfectly.

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Literature Review

Solar Powered Wheel Chair with Voice Controller for Physically Challenged Persons:

In recent times, assistive technologies have moved towards the inclusion of renewable energy and smart controls to provide better mobility solutions for physically challenged individuals. The following section presents a synthesis of work done previously related to this project:

1. Multimodal Control Systems

The early studies stressed voice and gesture controls to reduce reliance on manual operation. For instance, Haque et al. (2020) designed a solar-powered wheelchair controlled by gestures and voice, showing the potential of this technology to empower users with limited upper-body mobility. However, IoT was not integrated for health monitoring in the design. Madheswari et al. (2017) designed an Arduino microcontroller-based, low-cost voice-controlled wheelchair but without gesture control, hence limiting the flexibility of the users. These works highlight the need for hybrid control systems: voice, gesture, and manual, in order to meet various user requirements.

2. Integration of Solar Energy

Solar energy has cropped up as an environment-friendly alternative to grid-dependent systems. Vijayakumar et al. (2021) designed a solar wheelchair with smart features like fingerprint authentication and obstacle detection. Their prototype was economical but had problems in battery efficiency when sunlight is low. Correspondingly, Hans and Sandeep (2016) have also elaborated on the cost-effectiveness and eco-friendliness of solar panels but pointed out their dependency on weather conditions. These findings are in agreement with the current project, utilizing solar panels and lithium-ion batteries to approach energy reliability.

3. IoT and Health Monitoring

IoT-enabled health tracking is critical for real-time medical support. Baker et al. (2020) explored IoT's role in healthcare, advocating for platforms like ThingSpeak to monitor vital signs remotely. The current project builds on this by integrating heart rate and temperature sensors, enabling caregivers to track users' health—a feature absent in earlier models like Sakib and Mouri's (2017) low-cost wheelchair, which prioritized affordability over advanced telemetry.

4. Technical Implementation

Open-source frameworks like Arduino have democratized prototyping. Monk's (2016) guide on Arduino programming underpins many projects, including this wheelchair's motion control system. Practical tutorials (e.g., Adafruit's accelerometer guide) further validate the use of GY-61 sensors for gesture recognition. However, prior designs often lacked modularity, a gap addressed here through retrofitting capabilities for traditional wheelchairs.

5. Safety and Obstacle Avoidance

Safety remains the most critical existing gap in present systems. Though Chen (2018) summarized the potential in gesture recognition, he still pressed for collision avoidance sensors. Partially, infrared sensors have been implemented in this project, yet it recognizes several shortcomings in, for example, using ultrasonic sensors for distance. Earlier research, such as that by Haque et al. (2020), though detecting obstacles, did not conduct any robust tests in noisy conditions.

6. Cost and Accessibility

Affordability is one of the echoing factors. Sakib and Mouri (2017) worked on rural accessibility by designing a solar wheelchair for low-income users in Bangladesh. Their cost-benefit analysis proved that solar systems outperformed their grid-dependent alternatives. The project discussed here also follows the same philosophy, as it incorporates off-the-shelf components like HC-05 Bluetooth to provide major IoT features without increasing the price much.

Research Gaps Addressed

Hybrid Controls: Integrates voice, gesture, and manual inputs for versatility.

Integration of IoT: allows for real-time health monitoring, which was not available in previous solar wheelchairs.

Open-source Design: shares circuit diagram and code for community-based improvements.

Safety Enhancement: institutes infrared sensors, although not applied to the current prototype.

Pros:

Cool Voice + Gesture Controls: Allows users to choose whether to speak or wave at the wheelchair to make it move. Super futuristic and helpful for people who cannot use their hands much.

Solar Power = Free Energy: It uses sunlight to charge; therefore, it is not necessary to plug in. Ideal for sunny places or saving money on electricity.

Cheap to Build: Uses basic parts like Arduino and Bluetooth modules. It even says cheaper than buying a regular electric wheelchair.

Health Monitoring: Heart rate and temperature are tracked, so doctors can check up on users remotely.

Works with Old Wheelchairs: You can upgrade a regular wheelchair instead of buying a new

one.

Cons:

Bluetooth Range is Weak: Only works within 10 meters. If you go too far, the wheelchair won't listen to your phone.

Sunlight Addiction: Solar panels don't charge well on cloudy days. Bad weather-wheelchair might stop working.

No Crash Protection: No sensors to stop it from hitting walls or furniture. You gotta steer carefully!

Gestures Are Confusing: For older people especially, it can take time learning hand motions.

Battery Dies Fast: The utilization of Bluetooth coupled with the application causes quick power drain. Recharge all the time.

Student's Notes:

Add Safety Stuff: Add infrared or ultrasonic sensors to avoid collision.

Better Batteries?: Try to change the lead acid batteries to Li-ion. That would increase lifespan and speed up charging.

Real User Testing: Have real disabled users test it. Lab tests are cool, but real life is messy!

Code Share: Publish Arduino sketches online to allow others to bug fix or add features.

Backup Controls: If Bluetooth dies, the manual buttons must always work.

Rainy Day Plan: Add a plug-in charger for days when there's no sun.

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Literature Review

Design and Development of Solar-Powered, Voice-Automated and Gesture-Controlled Wheelchair with Obstacle Detection using Ultrasonic Sensor

The design of the intelligent wheelchair has evolved through several changes. This is informed by the need to ensure the creation of independence among people with various physical disabilities. The following are some of the progresses and limitations in assistive mobility technology concerning the subject of interest.

1. Voice and Gesture Controls Development

The early voice-operated wheelchairs relied on speech recognition systems that used simple commands such as "forward" or "stop". These kinds of systems made things more accessible but excluded the millions of people who, for whatever reason, speak inappropriately. Then came gesture-controlled systems which, with the use of accelerometers and gyroscopes, can translate hand motions into motion commands. It almost always required a wired connection or bulky hardware. Recent developments have targeted a wide range of user needs, from mute or visually impaired users, through the incorporation of voice and gesture controls together. In this project, this is furthered through the integration of wireless, MEMS-based gloves for gesture control, allowing users to toggle between modes using voice or hand gestures seamlessly.

2. Solar Energy Integration

Traditional electric wheelchairs are bound by grid power, thus making them less useful in remote areas. These issues called for the introduction of solar-powered designs whereby photovoltaic panels charge batteries. The early prototype models also had poor energy efficiency in cloudy conditions and slow charging. Today, contemporary systems make use of high-efficiency solar panels with lithium-ion batteries for better storage of energy. The project will apply these developments in conjunction with obstacle detection for continued mobility.

3. Obstacle Detection and Safety

Early wheelchairs had no safety features and thus posed the risk of collision in cluttered environments. Infrared sensors were introduced later to detect obstacles but had limited range and accuracy. Ultrasonic sensors took over from infrared

systems because they were able to detect objects up to 80 cm away, even in low light conditions. The project presented here uses ultrasonic sensors to initiate an emergency stop for the safety of visually impaired users.

4. Hybrid Control Systems

Modern assistive devices are increasingly integrating multiple technologies for better accessibility. For instance, some wheelchairs have integrated Bluetooth-enabled mobile apps for remote control, while others do not have gesture inputs. This work stands out by integrating voice commands, gesture controls, and manual switches into a single system. Users can toggle between modes-voice or gesture-based on their physical capability to help people with a wide range of disabilities.

5. Open Challenges

Weather Dependency: Solar panels work weakly on cloudy or rainy days and thus need backup charging mechanisms.

Cost Barriers: Advanced functionalities like MEMS gloves and IoT integrations are unaffordable for low-income users.

Limited Real-World Testing: Most of the prototypes are tested in labs and not tried with severely disabled persons.

Key Contributions of the Current Work

Multi-Modal Accessibility: Merges voice, gesture, and manual controls to serve users with diverse disabilities.

Improved Safety: Ultrasonic sensors avoid collision, and solar power provides the friendly eco-system.

Wireless Gesture Control: MEMS gloves with RF transmitter for intuitive hands-free navigation.

Cost-Effective Design: Designed using low-cost components like Arduino and Bluetooth modules to make the design scalable.

Future Directions

Future research probably will involve the incorporation of AI into adaptive navigation, wind/noise energy harvesting for charging, and cloud-based IoT for real-time health monitoring. It is overcoming the cost and weather limitations that will be very vital for mainstream adaptation.

Pros:

Voice + Gesture Controls = Super Flexible: Users can choose either voice or waving to move. Ideal for users who cannot use their hands or voice. According to some studies, hybrid systems like this are the future, such as Haque et al.'s 2020 paper.

Solar Power = Eco-Friendly: Charges using sunlight, so no electricity bills.
Vijayakumar's 2021 project showed solar works well but cloudy days are a problem.

Safety First with Sensors

Ultrasonic sensors avoid collisions. Kumar & Raja (2015)
also revealed that the sensors work better than infrared sensors.

Inexpensive to Make

Arduino and other basic materials are utilized. Madheswari et al. (2017) also came up with a wheelchair at low cost, but with gesture control.

Suitable for All

Mute? Use gestures. Can't move your hands? Use voice. This project includes all disabilities.

Cons:

Bluetooth Range Sucks

Works only within 10 meters. The signal is interfered with by walls. This even troubled Haque et al. in 2020.

Solar Panels Need Sun

Cloudy days mean wheelchair may die. According to Vijayakumar's team in 2021, this is a serious drawback.

No Fancy Sensors

Has only simple ultrasonic sensors. No LiDAR or cameras for smarter navigation, as with the 2022 AI model from Bakouri et al.

Expensive Gloves

MEMS gloves cost \$. Poor people cannot afford this-the 2017 paper by Sakib & Mouri warned about the costs.

Difficult to Learn

Gestures take practice. Older users might get confused; this was proved in the tests conducted by Madheswari et al. in 2017.

Student's Notes:

Add Better Sensors: LiDAR or cameras for smarter obstacle detection. In his study of 2022, Bakouri used AI for this!

Hybrid Charging: Solar + plug-in charger for rainy days. This, according to Vijayakumar's team in 2021, is not optional.

Real-Person Testing: Labs are fine; however, the real world is not clean. Sanders et al. in 2020 said that real-world testing does matter.

Share Code: Publish Arduino sketches online. According to Monk's 2016 book on Arduino, open-source promotes innovation.

Control System Backups: In case Bluetooth dies, buttons should work anyway. Always.

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Literature Review

Finger-Gesture Controlled Wheelchair with Enabling IoT:

The growth of wheelchairs controlled by gestures has increased within the last ten years due to the intervention of computer vision, IoT, and affordability. The synthesis of previous work and ideas in the field follows; this places the contribution of the proposed Finger-Gesture Controlled Wheelchair with Enabling IoT into context - Sadi et al. (2022):

1. Evolution of Gesture-Based Control Systems

Early gesture-controlled wheelchairs relied on wearable sensors or complex setups. For example, Kutbi et al. (2017) used egocentric cameras for head movement tracking but faced challenges with cost and user discomfort. Tejonidhi et al. (2017) employed eye-pupil tracking but struggled with real-time performance (70–90% accuracy). These systems often required users to wear devices, such as accelerometers or electrodes, which is not practical.

The proposed system provides a solution based on finger-gestures captured through a low-cost RGB camera and a CNN model. By getting rid of the needs of wearable sensors, this corresponds to the assertion of Gao et al. 2017, who studied hand gestures based on depth-of-view cameras without considering the economical aspects and disturbing background. Second, focusing only on finger manipulation rather than full-body or whole-hand-raising gestures makes systems more accessible by users with different mobility capabilities.

2. Integration of IoT and Safety

Obstacle detection and falling alerts are two major safety features in such assistive devices. Mahmud et al. (2019) proposed a system with integrated accelerometers and flex sensors, although requiring hardware to be worn on the body reduced comfort. This work proposes obstacle avoidance with ultrasonic sensors (50 cm range of view) and an accelerometer for fall detection, while introducing IoT-enabled SMS alerts via Google Firebase. This work follows the trend of IoT-based fall detection proposed by Islam et al. (2020), with extensions in real-time emergency messaging for users.

3. Cost-Effective Design

Most smart wheelchairs are too expensive to purchase. Desai et al. (2017) developed iris-controlled systems using MATLAB but faced high latency (5 seconds). Similarly, Oliver et al. (2019) used joystick manipulation with accelerometers, costing over 500. The proposed system reduces costs to under 300 by leveraging affordable hardware: Raspberry Pi 4 for CNN-based gesture recognition.

Arduino Uno for motor control.

USB RGB camera (1 MP) for finger tracking.

That corroborates with Bhuyain et al. (2019), a low-cost EOG-based wheelchair non-invasive in nature.

4. Computer Vision

The use of Haar Cascade classifiers for palm detection and KCF tracking for finger movement optimizes computational efficiency on low-power devices. The 2D CNN model achieves 97.14% testing accuracy with minimal training data (750 images), outperforming Lee & Ahn (2020)'s RGB-based action recognition systems. However, challenges persist in low-light conditions and rapid finger movements, echoing Shaik et al. (2015)'s findings on skin-color segmentation limitations.

5. Hybrid Control and User-Centric Design

The three-gesture control-stop, horn, drive-further simplifies the operation compared to Mahmud et al. (2019), which uses a multi-modal approach involving head/hand/eye tracking. The "Drive" mode will track finger directionality for intuitive navigation, hence minimizing the cognitive load for elderly or severely disabled users.

Innovations and Gaps Addressed

Affordability: Less than US\$300 by using off-the-shelf components.

Non-Invasive Control: Elimination of wearable sensors and joysticks.

IoT Safety: Real-time fall alerts and obstacle detection.

Simplified Gestures: Focus on finger movements rather than complex hand gestures.

Limitations:

Lighting conditions are necessary for gesture recognition.

Only three gestures can be used, limiting the possibility of advanced commands.

Lack of AI in adaptive navigation, such as path optimization.

Future Directions

Multi-Modal Integration: Add voice or eye-tracking for users with hand impairments.

Robustness: Improve gesture recognition in varying lighting using edge AI.

Energy Efficiency: Explore solar charging to complement the lead-acid battery.

Pros:

Super Cheap (Under \$300!)

Made from basic components such as Raspberry Pi, Arduino, and a low-cost camera, way cheaper than power wheelchairs found in very expensive commercial categories.

Finger Gestures = Easy Mode

Just flick your finger to move—and without any joysticks or sensors attached to your body. Perfect for those with weakened hands.

Lit Safety Features

Stops on its own in case it senses obstacles, like walls, and shoots emergency messages in case one falls. So intelligent!

97% Accurate Gestures

Rarely does the model mess up an "Stop", "Drive," or "Horn" order. Way stronger than old school systems.

Run on Raspberry Pi

No High-End-Laptop required - Raspberry Pi controls everything smoothly.

Cons:

Bad Lighting Ruins It

Skin detection works properly in good lighting conditions, performs poorly in rooms with poor lightning, and direct sunlight. Slow CameraEquals Laggy Movement

Only 30 FPS from the camera. You move your finger too fast, and it might freeze for a second.

Only 3 Gestures

No "reverse" or "turbo mode." You need to use "Drive" mode with finger directions, which feels kinda basic.

Battery is Heavy

Uses old-school lead-acid batteries. No solar option, so you gotta charge it all the time.

No Learning AI

Can't learn new gestures or adapt to your habits. It's stuck with the three commands forever.

Student Notes:

Solar Panels: Add Solar Panels, cut the weight of the battery and charge outside. Eco-friendly!

Improve the Camera: Fixing the lag in the camera could be done by a 60 FPS camera. It is a bit expensive, but will do.

Train More Gestures: Add "reverse" or "emergency stop." More data would be required for this, though.

Real-life Testing: Labs are great, but test in a busy park or shopping mall. Real life is messy!

Share the Code: Share Raspberry Pi scripts online. Let coders of the world improve it.

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Literature Review:

Solar Powered Wheel Chair: Mobility For Physically Challenged

Whereas the wheelchair started as a mere, simple, and manual device, because of the growing demands for the assurance of independence in persons with various physical disabilities, it has steadily transformed into one with sophisticated electric propulsion systems. The design and fabrication of a solar-powered wheelchair within the wider context of the emerging assistive mobility technologies is embraced by the paper to discern some emergent issues related to cost-effectiveness, sustainability, and adaptability (Gurram et al., 2012).

1. Historical Background and Conventional Wheelchairs

The early history of the wheelchair-from Farfler's 17th century self-propelled three-wheeled chair design to the first collapsible steel frame designed by Jennings and Everest in 1933-led into the modern times of mobility. Later, following World War II, electric wheelchairs came to prominence for use with war veterans, but the batteries, charged from the grid, along with other high costs, prevented universal use into the 1950s. For decades, conventional electric models have faced criticism for expensive repairs, as well as an overall limited lifespan of the battery and reliance upon wall outlets, Walton et al., 2010.

2. Electric and Solar-Powered Wheelchair Innovation

Recent innovation tried to transcend these:

Hybrid Systems : Further, included in the list were voice controlled wheelchairs Eishout, 2007 and also those controlled with gestures, although both remained very expensive to mass-produce.

Solar Integration:

The UAE's Haidar Taleb designed a solar wheelchair to show renewable energy in use; in production, it remained expensive.

Cost-Effective Designs: The research focused on low-cost designs made from locally sourced materials and simple electronics.

The work presented is in continuation with such efforts along the line of affordability and sustainability. The authors could present a prototype at ₹25,000 $\approx$ \$300 much cheaper than the commercially available models indigenously sourced from incepted frame made of cast iron to alloy wheels and attached 48V battery system charged by four solar panels of 18W each.

3. Major Innovations and Design Issues Solar Energy Conversion:

Solar panels reduce the grid dependence and ensure at least 4 continuous hours or 8 km range.

Series-connected panels successfully produce charging while a few vibration sensitivity tests came with their share of setbacks; slow recharging was also seen.

Mechanical Design

Hub Motors: Positioned within the wheels to provide direct drive by eliminating complicated transmission.

Customizable Frame: Users can have more accessibility with its modular height adjustment and steering mechanism.

Load Capacity: It is designed for 80–100 kg, with a speed limit of 3–5 km/h for safety.

Cost Optimization:

Locally sourced materials-for instance, cast iron, rubber casters-reduced costs.

Simplified electronics, such as charge indicators and a reverse horn, reached functionality versus affordability.

4. Limitations and Gaps in Research

This study, though adding much to accessible mobility, has the following limitations, including:

Environmental Conditions: Bad light reduces the efficiency of solar panels; strong vibrations shorten life.

Terrain Restrictions: It can only cover flat surfaces; hence, being very limited for outdoor use.

Battery Efficiency: Lead-acid batteries are heavy; lithium-ion usage may grant more lifetime.

Previous related research, such as that of Langbein and O'Connor using stationary wheelchair trainers from 1993-2002, referred to various points at which mechanisms of exercise or stair climbers could be applied - elements this development lacked.

5. Future Directions

Following are the avenues of further development listed by the authors:

Increased Solar capacity: Larger panel size or MPPT controllers to reduce charging time

Terrain Adaptability: All-terrain wheels or suspension systems for uneven terrain.

Better Battery Systems: Lighter and longer-lasting Lithium-ion batteries used for upgrade.

Pros:

Sensibly Affordable AF (₹25,000 $\approx$ \$300!)

Constructed out of local materials, such as cast iron and alloy wheels, this is pretty inexpensive-just about as cheap as most electric wheelchairs. Ideal to be

used in rural areas or by people with low incomes.

Solar Power = Eco-Friendly

Powered by four 18W solar panels, this does not need a plug-in from the wall. Save on electricity bills and work outdoors.

Uncomplicated Design

No high-end technology is used-just hub motors, simple steering, and a rust-proof frame. Easy to repair with locally available tools.

Adjustable Height

The solar frame can be raised/lowered. Great when users of varying height will be using it.

8 KM Range

Runs 4 hours on a full charge. Adequate for running errands or to the market.

Cons:

Slower Than a Tortoise

The top speed of 3-5 km/h makes pedestrians overtake you.

Sunlight Addict

Solar panels require direct sunlight. Cloudy days mean shorter trips.

Vibration Issues

Bumpy roads? Shake loose the solar panels-not good for rough terrain.

Heavy & Clunky

Made of cast iron, good to withstand but not that good to carry around.

Flat Surfaces Only

Cannot climb slopes, let alone stairs. No off-road capability whatsoever.

Student's Notes:

Better Batteries: Lithium-ion as opposed to lead-acid. Lighter, longer life, quicker charge.

Better Solar Panels: Avail either flexible or high-efficiency panels to reduce problems with vibration.

Suspension Addition: Shock absorbers for bad roads. What comfort!

Stair-Climbing Wheels: The future may have in store this design made by a guy from UAE by the name of Haidar Taleb in making a stair-climbing solar wheelchair.

Usability Testing with Real Users Let the disabled try it. Labs $\neq$ real life!

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Literature Review:

Sabertooth-Based Smart Electric Wheelchair with Advanced Features

The smart electric wheelchair has undergone rapid development in order to increase the autonomy, safety, and adaptability of users. The paper puts into context the novelties of the Sabertooth-Based Smart Electric Wheelchair as developed by Sohail & Saleem, in the forefront of innovations in assistive mobility technologies, contributing to motor control, health monitoring, and integration with

the IoT.

1. Motor Driver Advances

Traditional motor drivers, such as mechanical relays and solid-state devices (SSDs), face limitations in handling high currents and ensuring durability. Mechanical relays, while simple, are prone to damage from negative transients and lack precision in high-torque applications (Amin & Beard, 2016). SSDs, though faster, are vulnerable to peak currents in high-power systems. The Sabertooth motor driver addresses these issues by offering:

High Current Tolerance: Supports up to 60A, providing stable movement on slopes and ramps.

Regenerative Braking: Recovers energy during deceleration, improving battery efficiency.

Noise-Free Operation: Ultrasonic switching (24 kHz) eliminates audible interference.

This agrees with the attempts to enhance wheelchair motor reliability, as shown by Van Der Slikke et al. (2018), who emphasized torque management for terrain adaptability. The design of the Sabertooth outperforms earlier SSD-based systems, such as those proposed by Bourhis et al. (1993), that lacked integrated energy recovery.

2. Health Monitoring Integration

Clinician surveys indicate that 40% of wheelchair users face problems in steering and maneuvering the wheelchair, whereas 32% are unable to operate the conventional controls due to their poor motor skills. Ding & Cooper, 2005 The inclusion of pulse and temperature sensors in this project offers immediate, accurate health data; this is important for holistic patient care. Other similar health-centric designs include Pacnik et al. (2005), who developed voice-operated wheelchairs, but this system adds physiological monitoring—a feature increasingly demanded in smart healthcare frameworks (Attali & Pelisse, 2001).

3. IoT and Android App Control

The Android app of the wheelchair utilizes UDP to facilitate low-latency communication over Wi-Fi for easy control. In this regard, the IoT approach is similar to that of Devigne et al. (2020), where haptic feedback was provided for navigation. However, UDP, while efficient in real-time data transmission, does not utilize the error-checking protocols inherent in TCP, presenting some reliability trade-offs. Dual control—the joystick and touch-enhances accessibility, especially for those with limited dexterity, as noted in Posugade et al. (2012).

4. Obstacle Avoidance and Motor Calibration

The ultrasonic sensors for obstacle detection in the HC-SR04 extend the work of Murarka et al. (2008), who combined stereo and motion cues in safe navigation. Similarly, the encoders in this project make up for motor RPM discrepancies owing to manufacturing variabilities, guaranteeing movement balance that overcomes some of the weaknesses of McElvain et al. (2018) in purely neural circuits oblivious to mechanical discrepancies.

5. Experimental Validation

Testing demonstrated that the Sabertooth outperformed others on inclines, where most systems fail due to torque limitations. The encoder-adjusted RPM calibration ensures straight-line motion, addressing issues seen in Van Der Slikke et al. (2018). However, several wear-and-tear issues arise with the use of brushed DC motors, which could be a point of future work by using brushless alternatives.

Key Innovations and Future Directions

Motor Driver: The high current handling and regenerative braking of the Sabertooth set a new benchmark with respect to reliability.

Health Metrics: Integrates physiological monitoring, bridging mobility and healthcare.

IoT Flexibility: Android app control expands accessibility but warrants TCP integration for robustness.

Pros:

Sabertooth Motor Driver = Beast Mode

Handles 60A current, climbs ramps, and charges the battery during braking. No more burnt circuits! Inspired by Van Der Slikke's torque studies.

Health Monitoring

Pulse and temp sensors track your vitals. Docs love this-Ding & Cooper (2005) said health features are a game-changer.

Dual Controls (Joystick + Android App)

Use a joystick or tap your phone. Perfect if your hands are shaky. Similar touch controls were implemented by Posugade et al., 2012 .

Obstacle Detection

Ultrasonic sensors stop crashes. Murarka et al. (2008) used similar technology to ensure safe navigation .

Encoder Calibration

Wonky motor speeds get fixed automatically. No more driving in circles!

Cons:

Brushed Motors = Worn Out Fast

Brushed DC motors suck-they wear down. Should have used brushless like Rakasena's EMG wheelchair.

UDP Protocol = Risky

The app uses UDP; there are no error checks. If Wi-Fi drops, the wheelchair might ghost you. TCP would have been much safer.

Expensive AF

Sabertooth drivers cost \$. Not for broke college students.

Heavy & Bulky

All those sensors and motors add weight. Good luck lifting it into a car.

No AI Smarts

Missed chance to add self-driving mode. Bourhis et al. (1993) did automated navigation ages ago .

Student Notes:

Add Solar Panels: Cut battery dependency. Gurram et al. (2012) did it for ₹25k .

Swap to Brushless Motors: Longer lifespan, less maintenance.

Fix the App: Use TCP instead of UDP. Less lag, more reliability.

Lighter Materials: Aluminum frame? Carbon fiber? Make it portable!

AI Navigation: Copy Devigne's haptic feedback.

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LITERATURE REVIEW

SOLAR BASEDWHEEL CHAIR WITH ADVANCE FUNCTION

Yang et al. 2007 studied the application of motors at the rim of electric wheelchairs to lessen mechanical complexity but not torque. Earlier studies of wheelchair design focus on improving the comfort of users and energy efficiency. Their study shows that placement of motors is vital to balance the weight distribution that becomes crucial for stability while in motion. However, the design did not include smart sensors, hence has limited adaptability in cases of a user who has severe physical limitations.

Leary et al. (2012) carried out the biomechanical analysis of manual wheelchair propulsion. From the findings of their study, inappropriate seat height and handrim design caused muscle fatigue, especially for users who had long-term use. This study showed the demand for variable seating systems, and that fact later influenced our decision to include posture-customizable components in our design. Van der Woude et al. (2009) continued the research of the efficiency of submaximal propulsion, finding that lightweight materials and ergonomic handles could reduce energy expenditure by up to 20%. These insights guided our choice of aluminum alloys and foam-grip joysticks.

In this context of automation, Boninger et al. (1997) studied pushrim force dynamics and showed that due to the unequal distribution of force, repetitive strain injuries occur. This motivated our inclusion of obstacle-detection sensors-IR and ultrasonic-that can minimize abrupt stops and jerks. Vegter et al. (2013) later tested instrumented wheels on treadmills and proved that real-time feedback systems improve propulsion consistency. While their work focused on the use of manual wheelchairs, it motivated our use of RF communication for remote control and switch-based input.

Recently, there has been little attention given by earlier research work towards voice recognition and other controls based on EEG. For example, Janssen et al. (1994) have given more thought to the physical strain while ruling out the consideration of alternative ways of input in cases of poor mobility of the limbs, and the project will fill this gap in literature by integrating speech recognition with corneal-retinal potential sensing to allow control through either eye movements or voice commands as one sees fit and that are not possible with the previous models.

Sustainability issues are seldom a concern in the existing literature on wheelchair power sources. Most of them use lead-acid batteries, heavy and not very environment-friendly-issues also related to automotive batteries. Our solar-powered system develops further from Yang et al.'s motor efficiency principles by integrating renewable energy, hence reducing dependency on grid charging.

Gaps Provided by This Project:

Input Diversity: Integrating Joystick, Voice, EEG, and RF for broad accessibility.

Energy Sustainability: Solar panels complemented with lightweight Li-ion batteries replace conventional lead-acid systems.

Safety Features: IR/ultrasonic sensors for obstacle avoidance-addressing risks identified in biomechanical studies.

Pros:

Multi-Control Options: Works with joystick, voice commands, and eye movements (EEG), so even users with severe limb disabilities can operate it.

Solar Charging: Reduces dependency on grid electricity. Good for rural areas with power cuts!

IR Sensors: Safety sensors that automatically stop the chair in case there is a hole or an obstacle. Hopefully, no more falling downstairs.

Lightweight Design: Aluminum framing used along with a Li-ion battery lighter than heavy lead-acid makes it much easier to push manually.

Using Cheap RF Technology: HT12E/D encoder and decoder chips used are pretty low in cost compared to other Bluetooth/Wi-Fi systems.

Cons:

Expensive Sensors: EEG and photovoltaic panel =\$. Too expensive for everyone.

Complex Code: It was such a pain to code in C++ on the microcontroller. Took weeks to debug.

Battery Issue: Solar energy is not well applied on a cloudy day, and it takes 8-plus hours to back up the system by charging.

Weight Limit: Supports only up to 80 kg. Heavier users might break the frame; tested with my uncle... bad idea.

Student Notes :

EEG Stuff: Was trying to employ the use of corneal-retinal sensors for detecting winks. This worked in lab conditions, while outside lighting spoiled the readings. Had to install a filter-in, thanks, Prof. Rakesh!.

RF Glitches: Our dorm Wi-Fi used to interfere with the 433MHz signals and sometimes cause the chair to move on its own. Shielded the circuits with aluminum foil, and it was fixed.

Joystick Drama: The wires of the joystick would always break. Moved to a gaming console joystick -Xbox 360-much better!

Solar Panel Hack: ₹ 500 Cheap Amazon Panel, hacked. Once it got over heated, we put mini fan. Now it works.... Sort of.

Feedback of Class: The peer review says that is what it Looks like: "Sci-fi robot". Possibly a bit too futuristic for older elderly

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LITERATURE REVIEW

Wheelchair Control System based Eye Gaze

The evolution of eye-gaze-controlled wheelchairs is a function of developments related to assistive technology for people who have severe motor disabilities. Conventional electric wheelchairs depend on joysticks or voice, which is not feasible for users suffering from paralysis or speech-related disabilities (Kundu et al., 2018). Various recent works have been performed on alternate control mechanisms using EEG signals (Rashid et al., 2020) and eye tracking (Dahmani et al., 2020).

Key Achievements

Eye-tracking systems:

Early systems used infrared cameras and Haar cascades for pupil detection but faced challenges with calibration time and accuracy (Habet Tesfamikael et al., 2021). For example, [14] achieved 91% accuracy with a 4-directional wheelchair but struggled with vertical gaze detection.

Modern approaches leverage deep learning (dlib's 68-point facial landmarks) for real-time face and eye detection. The current study achieves 94.4% accuracy with a calibration time of 1 millisecond, outperforming earlier methods like [13] (90% accuracy, no calibration time reported).

Cost Efficiency:

Previous systems used expensive components like Raspberry Pi (70)(Singhetal.,2020).Thisworkemploysan**ArduinoUno(10)** and low-cost DC motors, reducing total costs to \$18, making it accessible for low-resource settings.

Multi-Directional Control:

Unlike older systems limited to 4 directions (left, right, forward, backward), this wheelchair enables 360° movement via gaze ratio calculations (sclera pixel analysis).

Pros:

High Accuracy: 94.4% is higher than that of older systems, such as 86% in [11].

Low Cost: \$18 due to the use of cheap components such as Arduino and L293D motor drivers.

Fast Calibration: Near-instantaneous detection via dlib's regression trees.

360° Mobility: Complex navigation compared to 4-directional systems.

Cons:

Lighting Dependency: The webcam-based system may not work well in low-light conditions.

Limited Vertical Gaze Detection: Focuses on horizontal eye movement (left/right/center).

Sclera Dependency: May fail for users with eye conditions reducing sclera visibility (e.g., cataracts).

No Obstacle Avoidance: Lacks ultrasonic sensors for collision prevention ([14] included this).

Student Notes :

- The thought of using eye movements to control wheel chairs is ingeniously thought of, allowing them to move around freely without having to depend on people. One possible issue could be that these systems still have technical limitations, mainly in tracking accuracy under variable lighting conditions.

- Some works show that it takes a really long time in calibration, and this can annoy the users. If researchers minimize this, the system will get more practical.
- Again, one of the best parts of this wheelchair is that there is no kind of physical strain involved, not like joystick ones. But over a time or so, after continuous use, the user would experience eye strain.
- Cost-wise, this technology is getting cheaper, but it is not yet widely available. If manufacturers can work on its affordability, many people will be able to avail themselves of this technology.
- Overall, the eye-controlled wheelchair is a brilliant invention, yet it still requires improvements to make it fully usable in daily life.

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